



Oguzhan Esen

📍 Munich, Bavaria, Germany ✉ ogguzhanesen@gmail.com ☎ +49 15258975738

📅 10.04.1999 🔗 <https://linkedin.com/in/oguzhanesen> 🐙 github.com/esenoguzhan

🚗 B

EDUCATION

Technical University of Munich 
Mechatronics and Robotics, Master of Science

10.2024 – 01.2027 | Munich, Germany

Yıldız Technical University
Electrical Engineering, Bachelor of Science
GPA: 3.57/4.0 (1.6 German equivalent)

2018 – 2022 | Istanbul, Turkey

WORK EXPERIENCE

Autonomous Vehicle Systems Lab, Technical University of Munich
Physical AI Research Assistant

07.2026 – Present | Munich, Germany

- Working with Unitree G1 humanoid, B2W quadruped, and AgileX robot arms
- Supporting SAP in leveraging large-scale business context data for robotics applications

AVL Software and Functions
Working Student

11.2024 – 07.2026 | Munich, Germany

- I have switched to a part-time role, continuing my responsibilities from my previous position after I started my master's.

AVL
Control Development Engineer

04.2022 – 10.2024 | Istanbul, Turkey

- Actively worked on model-based control software development projects for diverse applications including electric vehicles, hybrid systems, autonomous robotics, and e-mobility platforms in battery domain.
- Designed and implemented application-layer software using MATLAB/Simulink in accordance with ASPICE and MISRA compliance standards.
- Developed and maintained control functions and algorithms for BMS modules such as battery protection, thermal management, electrical hazard protection, contactor control, isolation measurement, cranking capability estimation, diagnostics, warranty management, thermal runaway detection, and vehicle communication.
- Executed Model-in-the-Loop (MiL) and Software-in-the-Loop (SiL) testing; actively supported Hardware-in-the-Loop (HiL) integration and validation processes.
- Requirement management and test management on component level
- Participated in the definition and enhancement of software architecture, ensuring modularity, reusability, and integration with broader system platforms.
- Developed a battery model calibration tool using Python and SciPy to optimize cell model parameters.
- Implemented deep learning algorithm prototypes for SOX estimation
- Collaborated closely with cross-functional teams from various disciplines and different AVL locations worldwide.

SIEMENS
Working Student

05.2021 – 04.2022 | Gebze, Turkey

- Supported after-sales operations and customer incident solutions.
- Troubleshooted non-conformities in medium and low voltage switchgears from both electrical and mechanical perspective.
- Coordinated different stakeholders and subject matter experts at different disciplines of operations.
- Ordered and managed critical manufacturing materials and procurement items.
- Organized and contributed to the material dispatch process and customs operations of materials.
- Tracked and reported continuously non-conformities of after sales scope in order to improve quality requirements.

THESIS WORK

Autonomous Aerial Systems Lab., Technical University of Munich 
Semester Thesis Student

10.2025 – 06.2026 | Munich, Germany

- Developed a safety-critical reinforcement learning framework integrating Proximal Policy Optimization (PPO) with Control Barrier Functions (CBFs) for safe quadrotor recovery and stabilization.
- Derived nonlinear quadrotor dynamics, Lie-derivative-based CBF constraints, and real-time safety-filtering pipelines for formal safety enforcement.
- Designed a curriculum learning strategy and reward shaping framework, combining dual-scale Cauchy rewards with goal-directed velocity objectives to improve recovery performance from arbitrary initial conditions.
- Implemented domain randomization, parameter uncertainty modeling, and calibrated disturbance injection to improve policy robustness and sim-to-real transfer.

- Extended the Flightmare simulation environment with Stable-Baselines3 integration, domain randomization utilities, and disturbance models for safe RL research.
- Designed and implemented a nonlinear Model Predictive Control (NMPC) baseline for comparative evaluation against the RL+CBF framework.
- Trained and evaluated controllers using PyTorch and Stable-Baselines3, demonstrating robust recovery from fully inverted states and large external disturbances.
- Deployed the learned policy on a real quadrotor (Agilicious), validating real-time performance and sim-to-real transfer.
- Results contributed to a research paper submitted to an international controls conference.

Autonomous Vehicle Systems Lab, Technical University of Munich

06.2026 – Present | Munich, Germany

Master Thesis Student

- Developing continual learning methods for multi-task robotic manipulation
- Investigating catastrophic forgetting and interference in flow-matching / diffusion policies
- Evaluating replay-free learning and compositional generalization across sequential manipulation tasks

MASTER PROJECTS

Foundational Models-based Autonomous Driving on F1TENTH

- Built a full ROS 2-based autonomous driving architecture for an F1TENTH vehicle deployed on NVIDIA Jetson Nano, integrating perception, localization, planning, and control modules.
- Integrated vision-language models (VLMs) and large language models (LLMs) for high-level decision-making and behavioral reasoning within a classical robotics pipeline.
- Implemented Stanley controller for stable lane tracking and path following.
- Designed and implemented finite state machine as a safety guard between low-level controller and foundational models.
- Conducted end-to-end system integration and testing in real-world scenarios on the F1TENTH platform, ensuring robust navigation and control performance under dynamic conditions.

Autonomous UAV Cave Exploration & Navigation

- Developed a ROS 2-based quadrotor system for 3D cave exploration in a Unity simulation environment. Implemented modular ROS 2 packages for mission control, enabling autonomous takeoff, state management, and mission execution. Integrated OctoMap for real-time 3D occupancy mapping and obstacle representation. Designed frontier-based exploration algorithms for unknown-space discovery and coverage, along with sampling-based motion planning for collision-free trajectory generation. Implemented a geometric SE(3) controller for full-state quadrotor control and stable tracking of aggressive 3D trajectories.

Mobile Robot Development for Intralogistics

- Developed an intralogistics LEGO robot system on a Raspberry Pi platform for warehouse navigation and logistics task execution. Implemented Dijkstra-based global path planning for efficient route computation and an OpenCV-based perception pipeline for object detection. Designed PID-based low-level motor control for a differential-drive mobile robot enabling stable trajectory tracking. Integrated MQTT-based communication for dynamic task allocation and human-robot interaction, achieving 1st place in the course challenge evaluation.

Humanoid Gait Imitation with Reinforcement Learning

- Developed a MuJoCo-based reinforcement learning framework for humanoid locomotion, incorporating OpenSim-based inverse kinematics from motion capture data for imitation learning. Designed reward shaping and training pipelines to enforce gait stability, motion tracking accuracy, and physically plausible locomotion.

Embedded Ball Control using PD controller on dsPIC33F

- Implemented a real-time PD control system on a dsPIC33F microcontroller for closed-loop circular trajectory tracking on a ball-and-plate platform. Developed low-level embedded software using registers, interrupts, ADC-based sensing, and PWM actuation for dual-servo motor control. The system was validated on real hardware, demonstrating stable circular trajectory tracking under real-time constraints.

PATENTS

Method and system for determining a charging current limit for a charging process of a rechargeable battery

- Contributed to a hybrid fast-charging framework combining physics-based electrochemical battery modeling with data-driven prediction and control for lithium-ion batteries.

SKILLS

Python

MATLAB&Simulink

PTC Windchill

C++

GIT

Object-Oriented Programming

Linux

ROS/ROS2

Codebeamer

LANGUAGES

English
C1, IELTS 7.0

Turkish
Native